STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-FINITE-PROJECTIVE-MODULES-TI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 429-439 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 441-473, the complete author proof (31 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 15 of queue_examples-defos.json (pb=1471, stmt 429-439, proof 441-473).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0D3I: sources/stacks-examples-defos/pinned-original-source.tex, source lines 359–369. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06JQ: sources/stacks-examples-defos/pinned-original-source.tex, source lines 473–473. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Example 0D3I let V be a finite dimensional k-vector space. Then*

$$T\mathcal{D}ef_V = (0) \quad \textit{and} \quad \mathit{Inf}(\mathcal{D}ef_V) = \mathit{End}_k(V)$$

are finite dimensional.

Proof. With $\mathcal{F}$ as in Example 0D3I set $x_0 = (k, V) \in \mathrm{Ob}(\mathcal{F}(k))$. Recall that $T\mathcal{D}ef_V = T_{x_0}\mathcal{F}$ is the set of isomorphism classes of pairs (x, α) consisting of an object x of $\mathcal{F}$ over the dual numbers $k[\epsilon]$ and a morphism $\alpha : x \to x_0$ of $\mathcal{F}$ lying over $k[\epsilon] \to k$.

Up to isomorphism, there is a unique pair (M, α) consisting of a finite projective module M over $k[\epsilon]$ and $k[\epsilon]$-linear map $\alpha : M \to V$ which induces an isomorphism $M \otimes_{k[\epsilon]} k \to V$. For example, if $V = k^{\oplus n}$, then we take $M = k[\epsilon]^{\oplus n}$ with the obvious map α.

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Similarly, $\mathrm{Inf}(\mathcal{D}ef_V) = \mathrm{Inf}_{x_0}(\mathcal{F})$ is the set of automorphisms of the trivial deformation x_0' of x_0 over $k[\epsilon]$. See Formal Deformation Theory, Definition 06JQ for details.

Given (M, α) as in the second paragraph, we see that an element of $\mathrm{Inf}_{x_0}(\mathcal{F})$ is an automorphism $\gamma : M \to M$ with $\gamma \bmod \epsilon = \mathrm{id}$. Then we can write $\gamma = \mathrm{id}_M + \epsilon\psi$ where $\psi : M/\epsilon M \to M/\epsilon M$ is k-linear. Using α we can think of ψ as an element of $\mathrm{End}_k(V)$ and this finishes the proof. $\square$